

CS441 Introduction to Computer Networking

.NET & C# Introduction

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What is .NET ?

- Microsoft .NET is a set of software technologies for connecting information, people, systems, and devices. This new generation of technology is based on Web services; small building-block applications that can connect to each other as well as to other, larger applications over the Internet.
- .NET is a development platform that makes it easier for programmers using different languages to quickly create robust internet applications.

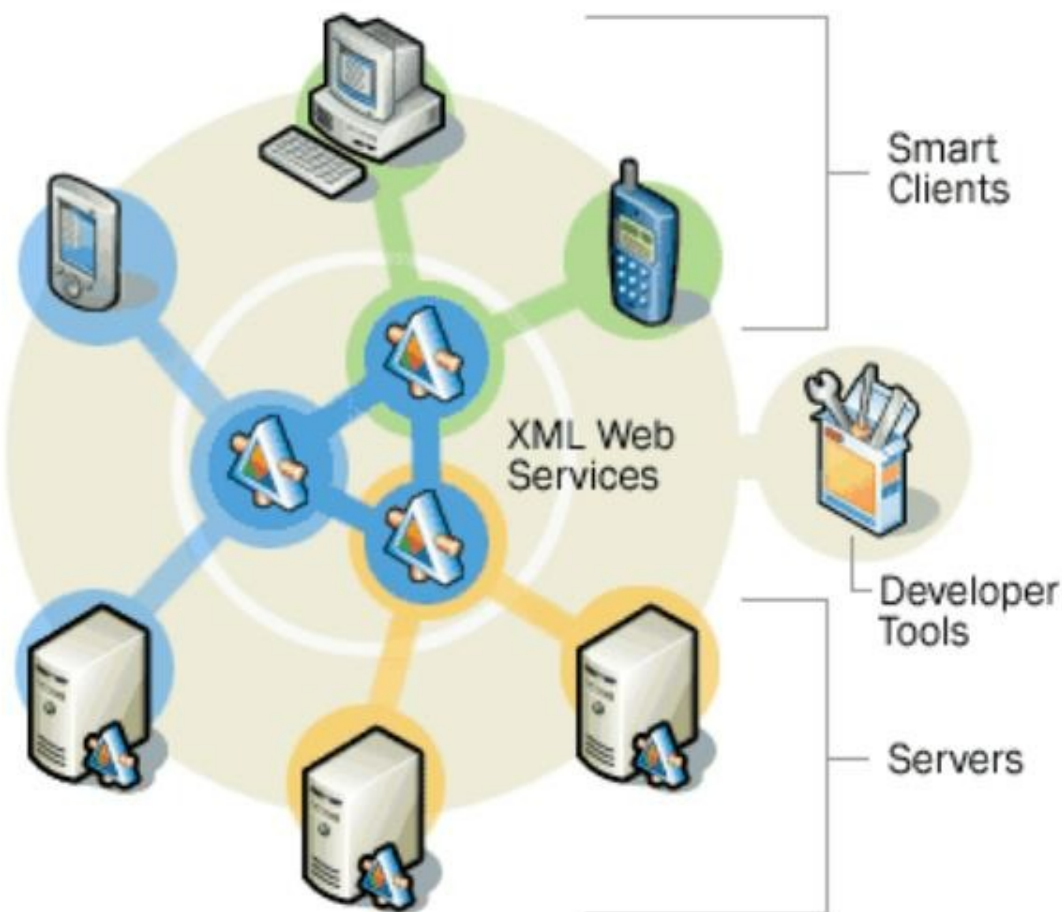
What is .NET ? cont.

- Language interoperability:

A developer picks the .NET language that he likes most, writes components in it, and shares the complied binary versions of his components with developers using other .NET languages

- .NET offers Base classes and Common Language Runtime

Defining the Basic Elements of .NET



Web services

are small, reusable applications written in XML, a universal language for data exchange. They allow data to be communicated across the Internet (or internal intranet) between otherwise unconnected sources that are enabled to host or act on them, for example:

Client-to-client: "Smart" clients or devices can host and apply XML Web services that allow data to be shared anywhere, any time.

Client-to-server: XML Web services can share data from a server application to a desktop or mobile computing device via the Internet.

Server-to-server: XML Web services provide a common interface between existing applications within an environment of independent servers.

Service-to-service: XML Web services can work together in sequence to create a more complex data operation.



Web Services

- Web services let applications share data, and -more powerfully - invoke capabilities from other applications without regard to how those applications were built, what operating system or platform they run on, and what devices are used to access them.
Although Web services remain independent of each other, they can loosely link themselves into a collaborating group that performs a particular task.

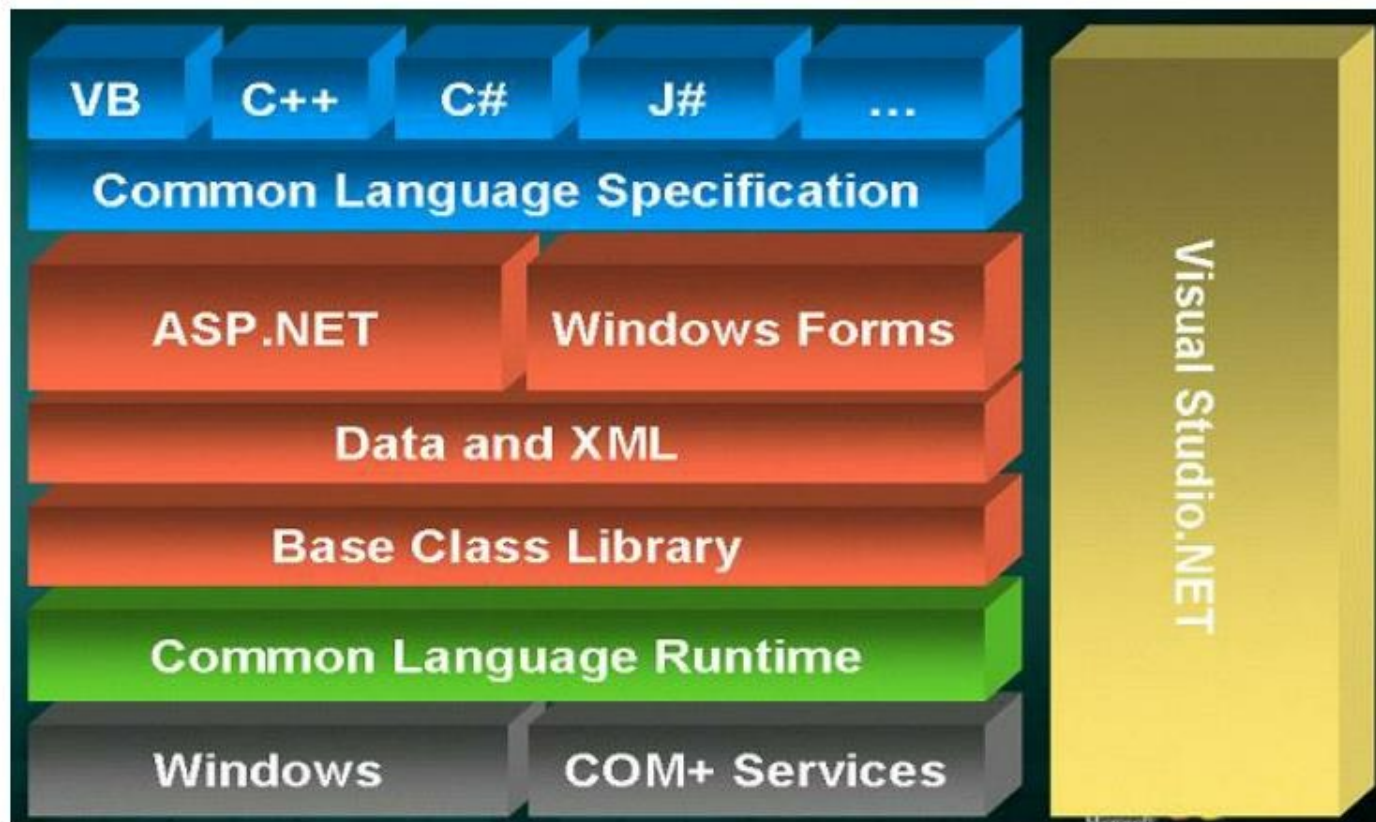
e.g. <http://www.microsoft.com/net/basics/webservicesoverview.asp>



What Is the .NET Framework?

- The .NET Framework is an integral Windows component for building and running the next generation of software applications and Web services.
 - Supports over 20 different programming languages.
 - Manages much of the plumbing involved in developing software, enabling developers to focus on the core business logic code.
 - Makes it easier than ever before to build, deploy, and administer secure, robust, and high-performing applications.
- The .NET Framework is composed of the common language runtime and a unified set of class libraries

.NET Framework & Development tool



Common Language Runtime

- The common language runtime (CLR) is responsible for run-time services such as language integration, security enforcement, and memory, process, and thread management.
- Its is similar to Virtual Machine of Java
 - Garbage collection
 - Common Type System (System.Int32, System.String, System.Boolean)
 - Just In Time Compile
 - Exception Handling
 - Security
 - Assembly

* Platform & Language Independence



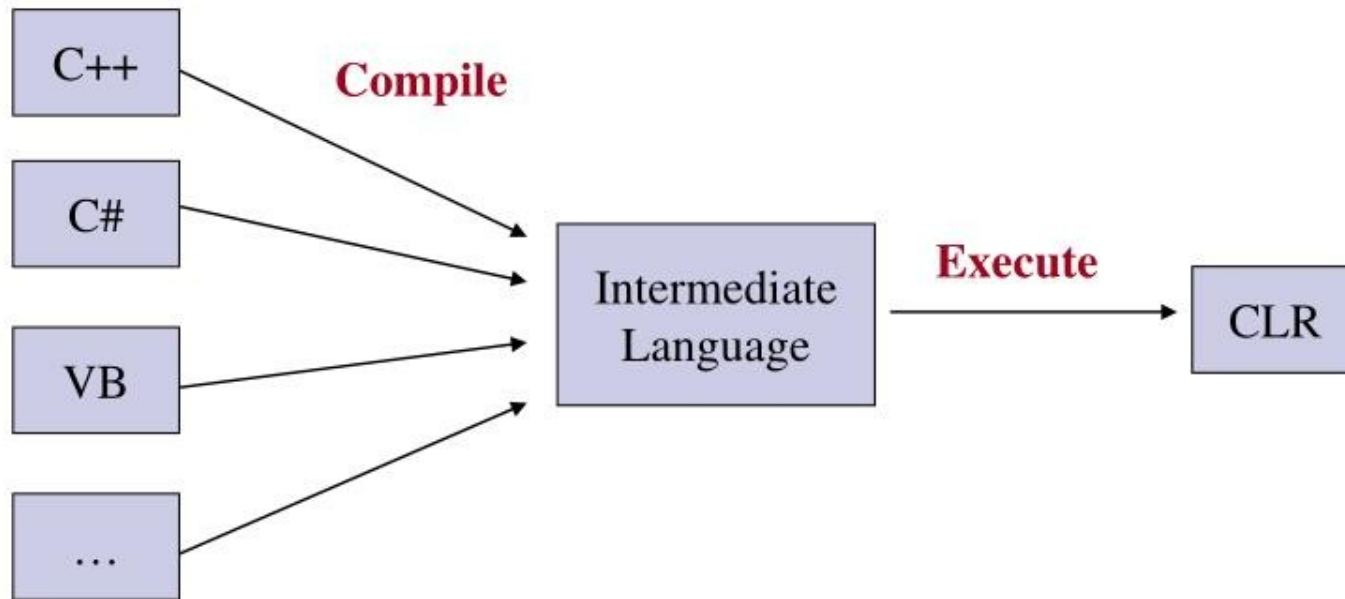
Runtime

- * Traditionally, a “runtime” was a component that a computer had to have in order to execute programs written in a particular programming language.

In order to execute Visual Basic programs, a machine had to have the VB runtime (msvbvm.dll) installed.

Java programs and C++ programs required unique runtime components too.

Intermediate Language (IL)



Language compatibility !!!



Base Class Libraries

- Base classes provide standard functionality such as input/output, string manipulation, security management, network communications, thread management, text management, and user interface design features.
- The ADO.NET classes enable developers to interact with data accessed in the form of XML through the OLE DB, ODBC, Oracle, and SQL Server interfaces. XML classes enable XML manipulation, searching, and translations. The ASP.NET classes support the development of Web-based applications and Web services. The Windows Forms classes support the development of desktop-based smart client applications.
- Together, the class libraries provide a common, consistent development interface across all languages supported by the .NET Framework.

.NET framework VS Java

☐ CLR	vs	JVM
☐ CTS, CIL	vs	JVMS
☐ Assembly	vs	Jar (Java Archive)
☐ Metadata	vs	Jar Manifest
☐ FCL	vs	Java APIs



Advantages of .NET

- Language Independence
- Better Support for Dynamic Web pages
 - ASP.NET
- Data Access
 - ADO.NET
- No More DLL Hell
- Improved Security
- Zero Impact Installation
- Support for Web Services



What is C#

- C# is Microsoft's new language, designed specifically for their new platform, the .NET Framework.
- C# is a simple, modern, object-oriented, and type-safe programming language that combines the high productivity of rapid application development languages with the raw power of C and C++



What to do with C#

- Console Applications
- Windows Applications
- Web Applications: ASP.NET
- Web Services

Socket

- A socket is a communication mechanism. A socket is normally identified by a small integer which may be called the socket descriptor. The socket mechanism was first introduced in the 4.2 BSD Unix system in 1983 in conjunction with the TCP/IP protocols that first appeared in the 4.1 BSD Unix system in late 1981.
- Formally a socket is defined by a group of four numbers, these are
 - The remote host identification number or address
 - The remote host port number
 - The local host identification number or address
 - The local host port number
- Users of Internet applications are normally aware of all except the local port number, this is allocated when connection is established and is almost entirely arbitrary unlike the well known port numbers associated with popular applications.
- To the application programmer the sockets mechanism is accessed via a number of functions.
 - `socket()` : create a socket
 - `bind()` : associate a socket with a network address
 - `connect()` : connect a socket to a remote network address
 - `listen()` : wait for incoming connection attempts
 - `accept()` : accept incoming connection attempts

```
using System;
using System.IO;
using System.Net;
using System.Text;
using System.Net.Sockets;

public class clnt {
    public static void Main() {
        try {
            TcpClient tcpclnt = new TcpClient();
            Console.WriteLine("Connecting.....");
        );
            tcpclnt.Connect("172.21.5.99",8001); // use the ipaddress as in the server program
            Console.WriteLine("Connected");
            Console.Write("Enter the string to be transmitted : ")
;
            String str=Console.ReadLine();
            Stream stm = tcpclnt.GetStream();
            ASCIIEncoding asen= new ASCIIEncoding();
            byte[] ba=asen.GetBytes(str);

            Console.WriteLine("Transmitting.....");
            stm.Write(ba,0,ba.Length);

            byte[] bb=new byte[100];
            int k=stm.Read(bb,0,100);

            for (int i=0;i<k;i++)
                Console.Write(Convert.ToChar(bb[i]));
            tcpclnt.Close();
        }
        catch (Exception e) {
            Console.WriteLine("Error..... " + e.StackTrace);
        }
    }
}
```

```

using System;
using System.Text;
using System.Net;
using System.Net.Sockets;

public class serv {
    public static void Main() {
        try {
            IPAddress ipAd = IPAddress.Parse("172.21.5.99"); // use local m/c IP address

            TcpListener myList=new TcpListener(ipAd,8001); /* Initializes the Listener */

            myList.Start(); /* Start Listeneting at the specified port */

            Console.WriteLine("The server is running at port 8001...");
            Console.WriteLine("The local End point is : " + myList.LocalEndPoint );
            Console.WriteLine("Waiting for a connection.....");

            Socket s=myList.AcceptSocket();
            Console.WriteLine("Connection accepted from "+s.RemoteEndPoint);

            byte[] b=new byte[100];
            int k=s.Receive(b);
            Console.WriteLine("Recieved...");
            for (int i=0;i<k;i++)
                Console.Write(Convert.ToChar(b[i]));

            ASCIIEncoding asen=new ASCIIEncoding();
            s.Send(asen.GetBytes("The string was recieved by the server."));
            Console.WriteLine("\nSent Acknowledgement");

            s.Close();
            myList.Stop();

        }
        catch (Exception e) {
            Console.WriteLine("Error..... " + e.StackTrace);
        }
    }
}

```



Reference

.NET - <http://www.microsoft.com/net/basics/>

.NET (korean) - <http://www.microsoft.com/korea/net/defined/default.asp>

S. Rovinson et al, Professional C#, 2nd Edition, WROX